RITESH KUMAR

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Professional Summary

Engineering student with a strong interest in Machine Learning, backed by solid web development skills. I've worked on building end-to-end AI applications, which has helped me develop a good understanding of machine learning concepts, NLP, and front-end design. I enjoy creating practical solutions by combining ML models with clean and interactive web interfaces. Along the way, I've taken part in hackathons, ideathons, and various college tech events, where I've improved my problem-solving abilities and learned how to work effectively in teams.

Education & Credentials

Bharati Vidyapeeth College of Engineering **2023-2027**
Computer Science Engineering
Angels Public School **2023**
CBSE Class XII

Professional Experience

GAIL INDIA LIMITED
MACHINE LEARNING INTERN *July 2025- Aug 2025*

- Engineered a production-ready House Price Prediction engine utilizing Linear, Ridge, and Lasso Regression algorithms.
- Architected robust data preprocessing pipelines encompassing missing value imputation, feature scaling, and outlier treatment.
- Used boston house price dataset to train and test the model.
- Boosted model accuracy to achieve an R^2>0.73 and reduced RMSE by 10% through hyperparameter tuning and EDA.
- Deployed the optimized machine learning workflow as an interactive, real-time web application using Streamlit.

Projects

Movie Recommendation System| *Python,Streamlit,Numpy,Pandas,Scikit learn*

- **Github :Movie Recommendation System**
- Engineered a comprehensive predictive analytics platform combining regression and classification algorithms to forecast academic trajectories and identify key performance indicators.
- Designed a predictive analytics pipeline to analyze and forecast academic outcomes using exploratory data analysis, statistical methods, and regression and classification techniques.

Customer Churn Prediction System| *Python,Scikit learn,Pandas,Numpy,JS,HTML,CSS,FastAPI,Matplotlib,Seaborn*

- **Github :Customer Churn Prediction System**
- Architected a classification-based ML framework utilizing Random Forest to predict attrition.
- Mitigated class imbalance to achieve 85%+ predictive accuracy to inform targeted retention strategies.

House Price Prediction System| *Python,Streamlit,Numpy,Pandas,Scikit learn,FastAPI,Docker*

- **Github :House Price Prediction Tool**
- Built an end-to-end regression-based machine learning solution using real-world housing datasets.
- Performed data cleaning, feature engineering, and exploratory analysis to identify high-impact pricing factors.
- Trained and evaluated Linear, Ridge, and Lasso models achieving $R^2 > 0.73$ with reduced RMSE, and deployed the solution using Streamlit for real-time user predictions.

Technical Skills

- **Programming Languages:** Python, SQL, C++, C, JAVA
- **AI & Machine Learning:** TensorFlow, Computer Vision & NLP, Scikit learn, NumPy
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Excel
- **Databases & Vector Stores:** PostgreSQL (Relational), MongoDB
- **Developer Tools & Cloud:** Git/GitHub, Docker, Streamlit , Google Cloud (Service Accounts), REST APIs
- **Core Engineering Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Software Engineering

Achievements & Extracurriculars

- **Awards**:
  1. 1st Place at Innovate and Automate(Flagship Event) Hackathon by ISA Delhi (Hardware).
  2. 4th Place & Specially Mentioned in Agentic AI Hackathon (Flagship Event) of IEEE(MAIT).
  3. Runner-up in 3 hackathons, Participated in 10+ Hackathons/Ideathons.
  4. **Administration Executive,** DSC BVCOE(**Sep 2024 - Present),**Venuva BVCOE(**Sep 2024 - Apr 2025**)